

# Beyond the Leaderboard: Counterfactual Diagnosis of End-to-End and VLA Driving Policies Under Domain Shift

Ruolin Yang, Zilin Huang, Buoyue Wang, Zhengyang Wan, Yuhao Luo, Zihao Sheng, Sikai Chen\*  
Paper ID —

**Abstract**—End-to-end and vision-language-action (VLA) driving policies are compared by leaderboard rank, but a deployment decision needs to know how a policy will behave at a new site, and why.

Attempting to predict a policy’s performance at a new deployment site from an existing leaderboard therefore works poorly, because a rank reflects only the outcome, not the behaviour behind it. On six released policies, rank on nuScenes open-loop metrics or on NAVSIM’s leaderboard does not carry over to scenes with a pedestrian near the ego corridor at a new site ( $\rho$  from  $-0.54$  to  $+0.49$ ).

We propose a counterfactual check-up: a few hundred real frames, each edited three ways (pedestrian removed, or re-rendered at dusk or at night), every edit verified by an independent detector, and the change in the planned trajectory read as a diagnosis rather than a score. By distilling causal feature axes from these edits, five exams separate what a score merges: how far the policy plans to drive, whether seeing the pedestrian buys safety, whether that response scales with danger, whether the plan moves when nothing requires it, and how much an irrelevant lighting change moves it.

Across 246 near-pedestrian scenes and six policies, among the cells in which the pedestrian does lie on the planned path only 1.9% of responses are genuine avoidance, and on those cells, under our open-loop protocol, no policy’s plan diverts or slows. In a pre-registered test from left- to right-hand drive, orderings and verdicts transfer but point values do not. The exam whose ordering best matches how often a policy keeps clear of the pedestrian on the new side is how steadily it drives ( $\rho = +0.94$ ), not how strongly it reacts. Most verdicts settle within a few dozen frames, while hazard sensitivity needs hundreds. Code and edited frames will be released.

## I. Introduction

A leaderboard rank does not tell you how a driving policy will behave in a new domain. End-to-end and vision-language-action (VLA) driving policies are compared by benchmark scores, and a deployment decision is, implicitly, a bet that the rank those scores produce will still hold once the policy is driving at the new site, not just on someone else’s roads. On scenes where a pedestrian is near the ego corridor, that bet fails for the six released policies we test: their rank on nuScenes’ [1] open-loop metrics or on NAVSIM’s [2], [3] leaderboard score does not carry over to the deployment site ( $\rho = -0.54$  to  $+0.49$ , Section IV-E), a gap that open-loop scores are known to open [4]–[6]. The reason is not noise but mechanism. A score is an outcome measure, and an outcome can be right for the wrong reason: a policy that never approaches a pedestrian

because it plans short trajectories scores as well as one that sees the pedestrian and yields, and the two behave very differently the first time the scene changes. Telling them apart means asking why the score came out as it did, usually from little data, because a closed-loop simulator for the target domain or a fleet test is exactly what the decision is meant to save.

However, no comprehensive evaluation method yet measures systematically how a policy will perform in the target domain, so organisations deploying a model across domains fall back on extensive testing and post-training in the deployment setting itself—rider-only fleets accumulating tens of millions of miles before a safety claim can be made [7], and a generalist robot policy improved through reinforcement learning on its own deployments and on-robot data collection [8]—a loop whose cost, measured in real-world miles, is prohibitive [9]. This makes selecting a model for cross-domain deployment harder still.

A considerable body of cross-domain work has been proposed—domain adaptation [10], domain generalisation [11], [12] and the like—but it concentrates on identifying the problem and on solving it through training, and lacks any systematic assessment of how a policy behaves in the target domain. It therefore does not substantively address the selection problem that arises at deployment.

Closer to our setting, [13] improves a VLA’s out-of-distribution generalisation by aligning its visual representations, but it presumes an already-chosen vision-language model and offers no procedure for selecting among policies, vision-only ones included. Representation engineering [14] locates and steers concepts in a language model’s activations, yet it is built around semantic prompts and does not carry over to policies whose input is images. Function vectors [15] show that a task can be read out of, and transferred through, a language model’s hidden states. The same instruments have no counterpart for the cross-domain evaluation of physical-AI policies. A systematic method for that evaluation is still missing.

To address this, we propose the Check-Up, a counterfactual diagnosis in three stages (Fig. 1). (a) We construct pixel-level paired data: every logged frame is queried with and without the target pedestrian and under a night-style photometric perturbation at two levels, and an independent detector verifies every edit. (b) From

each pair we distil two axes of causal attribution—the faithfulness displacement  $F$ , the plan change caused by the pedestrian itself, and the interference displacement  $I$ , the change the same policy produces under an edit that carries no driving-relevant information (3). (c) Five exams, each with a control and a reference threshold, are built on these paired plans and axes (Table I): how far the policy plans to drive, whether seeing the pedestrian buys safety, whether that benefit scales with the hazard, whether the plan moves when nothing requires it, and how much the irrelevant edit moves it. (d) Read on one driving side and tested on the other, the exams yield a prognosis for the target domain: the exam that transfers reproduces the new side’s ordering by how often a policy keeps clear of the pedestrian ( $\rho = +0.94$ ), where the standard scores reach only from  $-0.54$  to  $+0.49$ , and each verdict can be priced in frames before it is used.

The contribution is the Check-Up, a counterfactual diagnosis of end-to-end and VLA driving policies under domain shift, in three parts.

- 1) Pixel-level paired data. Every logged frame is queried with and without the target pedestrian and under a night-style perturbation at two levels, each edit verified by an independent detector. Six released policies are read on two driving sides.
- 2) Two causal axes. A response is read as the faithfulness displacement  $F$  and the interference displacement  $I$  (3), and counts as avoidance only if it moves, exceeds the jitter, and buys distance. Where the pedestrian lies on the planned path, 1.9% of responses qualify, and the lighting edit moves every plan further than the pedestrian does.
- 3) A priced prognosis. Diagnosed on one driving side and tested on the other, orderings and verdicts transfer: the exam that transfers reproduces the new side’s clearance ordering ( $\rho = +0.94$ ), against  $-0.54$  to  $+0.49$  for the standard scores, and each verdict is priced in frames—lighting within 14, exposure within 51.

## II. Related Work

a) Driving policies and their benchmarks: We examine bird’s-eye-view (BEV) planners of the TransFuser lineage [16] (LTF, DiffusionDrive [17], DiffusionDriveV2 [18]) and vision–language–action models (SimLingo [19], AutoVLA [20], Alpamayo-1.5 [21], [22]). NAVSIM [3], [23] scores their plans by multiplying hard gates with a weighted average of soft terms. Open-loop scores are known to reward proxies [5], [24], and closed-loop benchmarks [25] score a policy in simulation, whereas we read released policies on real frames. We show that the six-policy spread is carried by a single gate.

b) Behavioural testing: Checklists of targeted behavioural tests [26] and corruption benchmarks [27]–[29] probe models beyond aggregate accuracy. A metric drop,

however, does not say whether a policy became more cautious or less. Our exams read signed plan changes against a counterfactual and state how many frames each needs.

c) Causal confusion: Imitation learners can respond to correlates of the expert’s action rather than to its causes [6], [30]–[32]. Our exams apply this idea to trained policies: a cause (a pedestrian) and a nuisance (lighting) are intervened on directly.

d) Counterfactual editing and internal representations: Counterfactual explanations edit the scene [33], [34], and CausalAgents [35] deletes non-causal agents to test motion forecasters. Our removal edit applies the same idea at pixel level to a driving policy’s input, and we verify every edit with a detector. Steering directions [14] can show that a network encodes what it does not act on [36]. We use them to explain a diagnosis, not as the diagnosis.

## III. The Check-Up: Interventions, Axes and Exams

### A. Units, interventions and plans

Write a policy as a map  $\pi$  from a sensory-and-state input to a planned trajectory  $X = (X_t)_{t \in [0, 2.5]}$ ,  $X_t \in \mathbb{R}^2$  in the ego frame, sampled at 0.1 s. A unit is a pair  $u = (f, v)$ : a logged frame  $f$  and an input ego speed  $v$ . We take  $v$  from the log for every frame and, wherever the pedestrian’s logged future is available, additionally  $v \in \{4, 8\}$  m/s, which places the pedestrian on the planned path in scenes where the logged ego was stationary. The same grid is used on both corpora so that all six policies are read on one unit pool. The readouts are insensitive to it (widening the grid to  $\{2, 4, 6, 8\}$  leaves the specificity and scaling orderings identical,  $\rho = 1.00$ ).

Three interventions act on the input, leaving the policy untouched:

$$\begin{aligned} \mathcal{E}_O &= \text{id}, & \mathcal{E}_R &: \text{delete the pedestrian}, \\ \mathcal{E}_D, \mathcal{E}_N &: \text{re-light at dusk and at night}, \end{aligned} \quad (1)$$

and yield the paired plans  $X^O = \pi(\mathcal{E}_O u)$ ,  $X^R = \pi(\mathcal{E}_R u)$ ,  $X^D = \pi(\mathcal{E}_D u)$ ,  $X^N = \pi(\mathcal{E}_N u)$ . The letters  $O, R, D, N$  name the four conditions only: an intervention carries one as a subscript, a plan as a superscript. All quantities below are functionals of these plans and of the pedestrian’s logged future  $P_t$ . The horizon 2.5 s is the one all six policies share.

The displacement between two plans is

$$\bar{D}(X, Y) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|X_t - Y_t\|, \quad \mathcal{T} = \{0, 0.1, \dots, 2.5\}, \quad (2)$$

and the two quantities the rest of the paper is written in are

$$F = \bar{D}(X^O, X^R), \quad I = \bar{D}(X^O, X^N). \quad (3)$$

We call  $F$  the faithfulness displacement—the plan change attributable to the pedestrian itself—and  $I$  the interference displacement—the change the same policy produces under an edit that carries no driving-relevant information, and therefore its own jitter floor.  $I$  is taken at the night

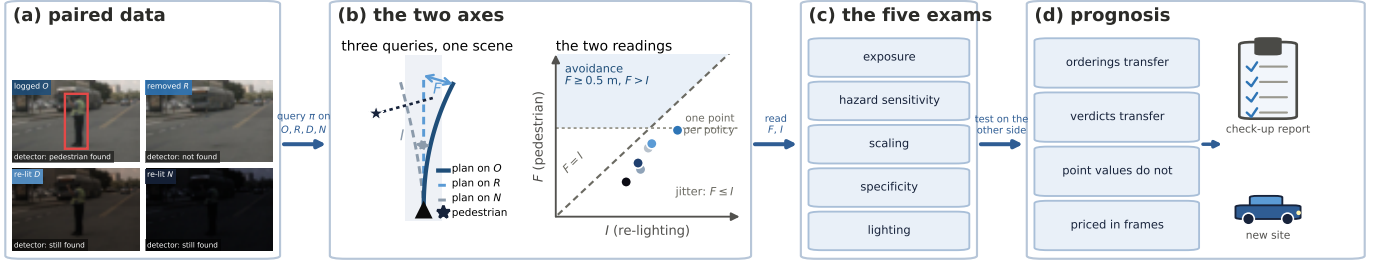


Fig. 1. The Check-Up framework. Pixel-level paired data (a) yield the two causal axes (b) and the five exams built on them (c). Diagnosed on one driving side and tested on the other, they give a priced prognosis, the check-up report (d).

TABLE I

The check-up. Each exam: its question, measurement, the plans or axes of Section III it is built on, and the reference threshold we set.

| Exam               | Question                   | Measurement                                         | Built on               | Ref.                        |
|--------------------|----------------------------|-----------------------------------------------------|------------------------|-----------------------------|
| Exposure           | How far ahead it commits   | planned over logged distance, 2.5 s                 | $X^O, H$               | $\approx 1$                 |
| Hazard sensitivity | Does it react to danger    | safety gained by seeing the pedestrian, when needed | $X^O : X^R$            | $\rightarrow 1$ with hazard |
| Scaling            | Does caution scale with it | rank corr. of HS and hazard                         | HS vs $a_{\text{req}}$ | $> 0$                       |
| Specificity        | Does it over-react         | plan change when no reaction is needed              | $F$                    | $\approx 1$                 |
| Lighting           | Does it react to noise     | perturbation vs. removal, same frame                | $F/I$                  | CFR $\gg 1$                 |

level. The dusk reading  $\bar{D}(X^O, X^D)$  enters Section IV-D. A policy that uses the pedestrian and ignores the lighting has  $F \gg I$ . The reverse ordering means the plan is driven by something the road does not care about.

a) Realising  $\mathcal{E}_R$ : The target is localised by projecting its annotated 3D box into the camera rather than by taking the dataset’s 2D boxes, whose association to the tracked object is unreliable for the short-range, partially occluded pedestrians of these scenes. It is segmented with SAM [37] (Fig. 1a) (box and torso prompts, clipped to the box, dilated by 9 px) and inpainted with LaMa [38]. For the LiDAR policy the points inside the box are deleted. Vision-language-action (VLA) policies read several cameras and only the front one is edited, so we retain only frames in which the target appears in no other view.

b) Realising  $\mathcal{E}_D$  and  $\mathcal{E}_N$ : Each image  $\mathbf{f} \in [0, 1]^{h \times w \times 3}$  the policy consumes is mapped by a closed-form photometric transform

$$\mathcal{E}_{D,N}(\mathbf{f}) = \text{clip}[g(\text{sat}_s \mathbf{f})^\gamma \odot \boldsymbol{\tau} + \sigma \epsilon], \quad \epsilon \sim \mathcal{N}(0, 1), \quad (4)$$

with  $\text{sat}_s$  the saturation scaling,  $\boldsymbol{\tau}$  a per-channel tint and the noise seed fixed per frame. This is a photometric night-style perturbation (gamma, gain, saturation, noise), not a captured night scene, and we refer to it as the night-style perturbation. Two levels are used:  $\mathcal{E}_D$ , the milder dusk  $(\gamma, g, s, \sigma) = (1.45, 0.70, 0.80, 2/255)$  with a warm tint, chosen to keep the pedestrian visible, and  $\mathcal{E}_N$ , night  $(2.2, 0.45, 0.45, 7/255)$  with a blue tint (Fig. 1a, Section IV-D).

c) Validity of the interventions: An independent detector (Faster R-CNN [39], COCO, score  $\geq 0.5$ ) is run on the 248 front-only frames. It recovers the target in 100% of the original images, in 8.5% under  $\mathcal{E}_R$  and in 89.9% under  $\mathcal{E}_N$  (IoU  $\geq 0.5$ ):  $\mathcal{E}_R$  suppresses the pedestrian for an observer outside the policy, while  $\mathcal{E}_N$  leaves it

visible and therefore changes the lighting rather than the scene. Two controls bound the residual artefact: pasting inpainting patches on empty road puts it at 5–17% of the removal effect at pedestrian scale, and removing off-road pedestrians, which should not matter causally, gives a same-type null.

## B. Readouts

With  $c_t(X) = \|X_t - P_t\| - r_e - r_p$  the clearance net of the ego half-width  $r_e = 1.0$  m and the pedestrian radius  $r_p = 0.4$  m, a plan is summarised by four functionals,

$$S(X) = \text{mean}_t \min(c_t, \bar{c}), \quad A(X) = \mathbb{K}[\min_t c_t > 0], \quad (5)$$

$$C(X) = \text{clip}(\min_t c_t / d_0, 0, 1), \quad T(X) = \frac{1}{2.5} \inf\{t: c_t < d_0\}, \quad (6)$$

with  $T = 1$  if the set is empty. The constants are physical, not tuned:  $r_e$  is the half-width of the nuScenes ego box ( $4.08 \times 1.85$  m),  $r_p$  the disc a standing adult occupies,  $d_0 = 1$  m the outer edge of personal space [40], and  $\bar{c} = 10$  m caps the contribution of scenes in which the pedestrian is far away. Equations (5)–(6) instantiate the standard surrogate-safety measures—contact, minimum clearance and an encroachment-time variant [41], [42]—read as distance to a person over the whole trajectory, as is conventional for navigation near people [43]. Speed is deliberately not a readout: the same deceleration is safe or unsafe depending on where the person is.

## C. Exams

Let  $L(X) = \sum_t \|X_{t+\delta} - X_t\|$  denote arc length and  $H$  the logged human trajectory.

a) Exposure:  $\text{Exp} = \text{med } L(X^O) / L(H)$  over frames with  $L(H) \geq 0.5$  m.

b) Hazard sensitivity: A unit needs a reaction when the pedestrian-blind plan enters personal space,  $C(X^R) < 1$ . On those units, writing  $\Delta(\cdot) = (\cdot)(X^O) - (\cdot)(X^R)$ ,

$$\text{HS} = \frac{1}{4} [\Delta A + \Delta C + \Delta T + \tanh(\Delta S/d_0)] \in [-1, 1]. \quad (7)$$

Each term lies in  $[-1, 1]$  by construction and  $\tanh$  squashes the single unbounded one at the  $d_0$  scale. The composite is ours, its ingredients are not. Its four terms are the surrogate-safety measures of (5)–(6), and averaging normalised sub-scores into one index is how the benchmark we compare against forms its extended predictive driver model score (EPDMS) [3], keeping the two commensurable. What the check-up adds is that every sub-score is a paired difference across  $\mathcal{E}_R$ —the ablation form used to attribute behaviour to an input in interpretability work [14], [44]. HS is read against the model-independent hazard

$$a_{\text{req}} = \frac{v^2}{2 \max(d-2, 0.5)}, \quad \text{TTC}_0 = \frac{d-2}{v}, \quad (8)$$

the deceleration required to stop short of a pedestrian at distance  $d$  and the corresponding time to collision (TTC) [41], [42]. Units with  $v < 1$  m/s are excluded: a stationary ego can move neither toward nor away from anyone.

c) Scaling:  $\text{Sc} = \rho(\text{HS}, a_{\text{req}})$  over needed units, with  $\rho$  Spearman’s rank correlation.

d) Specificity: On units needing no reaction, the minimal-change principle gives

$$\text{SP} = \exp(-\bar{D}(X^O, X^R)/0.5 \text{ m}) \in (0, 1], \quad (9)$$

so  $\text{SP} \rightarrow 1$  iff an irrelevant edit leaves the plan where it was.

e) Lighting: The causal faithfulness ratio (CFR) weighs the two displacements of (3) against each other,

$$\text{CFR} = \frac{\mathbb{E} \bar{D}(X^O, X^R)}{\mathbb{E} \bar{D}(X^O, X^N)} = \frac{\mathbb{E} F}{\mathbb{E} I}, \quad (10)$$

A policy for which lighting is irrelevant has  $\text{CFR} \gg 1$ . The thresholds of Table I ( $\text{HS} \geq 0.15$ ,  $\text{Sc} > 0$ ,  $\text{SP} \geq 0.8$ , exposure within  $[0.8, 1.2]$ ,  $\text{CFR} > 1$ ) are reference thresholds we set, not population norms.

## IV. Experiments

### A. Setup

We examine LTF and DiffusionDrive (front camera and ego status), DiffusionDriveV2 (camera and LiDAR), SimLingo (InternVL2-1B), AutoVLA (Qwen2.5-VL-3B, three cameras  $\times$  four frames) and Alpamayo-1.5 (four cameras, a diffusion head over the vision–language model (VLM) cache), all with released weights. The three BEV planners are trained on NAVSIM (nuPlan) data, as is AutoVLA’s released checkpoint. SimLingo is trained in CARLA and Alpamayo-1.5 on NVIDIA’s PhysicalAI-AV recordings. None is fine-tuned here, so every policy is read on a corpus and sensor rig it was not trained on. For the

BEV planners the nuScenes front image is centre-cropped to the 4:1 aspect of NAVSIM’s stitched front view with the sky removed and resized to the network’s input size, the logged speed and acceleration fill the status vector, and DiffusionDriveV2 receives the LiDAR sweep. The VLA policies receive the nuScenes cameras mapped approximately onto the views they expect. DiffusionDrive’s diffusion seed is fixed so that paired queries share their noise.

The primary corpus is nuScenes [1] keyframes: corridor pedestrians (lateral clearance to the ego corridor  $\leq 1$  m) at 5–12 m, their off-road counterparts, and 30 approach sequences at 5.8–14.6 m with logged pedestrian futures. The near-pedestrian units come from 236 front-only frames, 88 in Boston (left-hand drive) and 148 in Singapore (right-hand drive). Alpamayo-1.5 covers 170 of the 236, needing 1.6 s of history and 6.4 s of future within the scene. Section IV-B repeats  $\mathcal{E}_R$ ,  $\mathcal{E}_D$  and  $\mathcal{E}_N$  on a second, right-hand-drive corpus—the NAVSIM test split [3], 254 Singapore scenes with a pedestrian near the ego corridor, of which 246 carry a complete logged pedestrian future—used only for the transfer and decomposition results. Table II is measured on the nuScenes frames alone.

### B. The Two Axes

The diagnosis reads a policy on the two axes of (3). We apply both interventions to the 246 NAVSIM Singapore near-pedestrian scenes and read, per scene,  $F$ , the plan displacement caused by removing the target pedestrian, and  $I$ , that caused by the night-style perturbation.

a) When the plan does move, which way does it move?:  $F$  measures how far a plan moves, not where it moves to. We therefore decompose it (Fig. 2d), reading  $F$  against  $I$  of (3)—the same policy’s jitter floor under an unrelated edit—and against  $\Delta S$ , the change in clearance to the pedestrian’s logged future. A response counts as avoidance only if all three hold: the plan moved ( $F \geq 0.5$  m), it moved more than the policy’s own jitter ( $F > I$ ), and the movement bought distance ( $\Delta S \geq 0.5$  m).

Three readings follow. First, the pedestrian outweighs the irrelevant edit in only 4–23% of scenes, so most of what  $F$  registers sits inside the policy’s own variability—below the  $F = I$  line of Fig. 2d. Second, avoidance is rare where it is called for. The logged ego is stationary through most of this corpus, so a plan that does not move is usually the correct one. We therefore replay every scene at input speeds of 4 and 8 m/s, as in Section III, and read only the cells in which the pedestrian-blind plan does come within 1 m of the pedestrian’s logged future. Across 6 policies that is 787 of 4428 cells (11–30% per policy), and 15 of them—1.9%—are genuine avoidance. On those cells the median clearance changes by  $+0.00$ – $+0.03$  m and the median planned distance by  $-0.02$ – $+0.08$  m: facing a pedestrian it would otherwise reach, no policy’s plan diverts or slows under our open-loop protocol. Third, the displacement that does occur is not aimed: of the 121

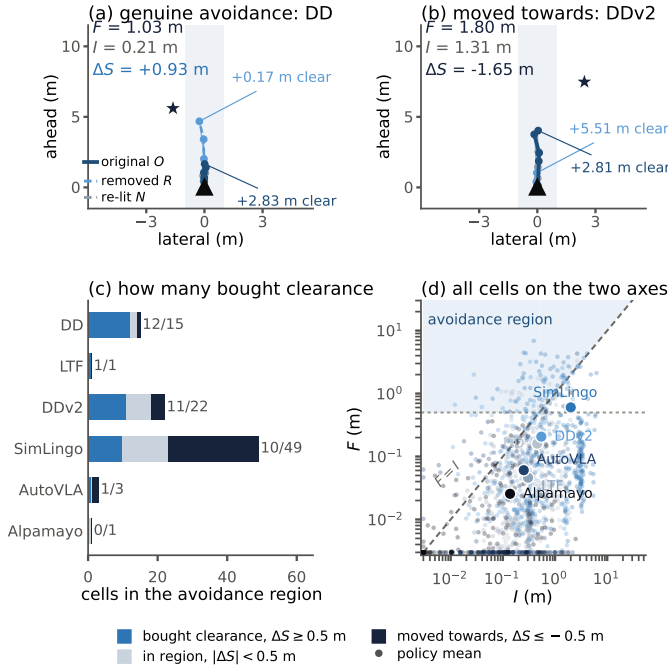


Fig. 2. The two axes. (a) Genuine avoidance: with the pedestrian in view ( $X^O$ , solid) DiffusionDrive stays short of its logged future (dashed from the star), while blind to it ( $X^R$ , dashed) it reaches it.  $X^N$  gives  $I$ . (b) Moved towards: DiffusionDriveV2 plans further with the pedestrian than without. (c) Cells in the avoidance region ( $F \geq 0.5$  m,  $F > I$ ) per policy, by  $\Delta S$ : green  $\geq 0.5$  m, red  $\leq -0.5$  m, grey between. Label: green over all. (d) All 246 Singapore cells on the axes of (3), by policy, with means.

cells in which the plan moved, 43 increased the clearance and 42 decreased it. Per policy the split differs sharply (Fig. 2c): DiffusionDrive reacts in few scenes but in the right direction (median  $\Delta S = +2.16$  m), while SimLingo moves in more scenes than the other five combined and moves towards the pedestrian more often than away (32 against 13, median  $-0.59$  m), and it also has the highest rate of sub-1.5 s time-to-proximity (Table V). SimLingo is trained in simulation and run here on real images, so part of what we measure is a sim-to-real gap, which is exactly what a check-up at a new site should surface. Finally, what a policy yields does not grow with the hazard: the rank correlation between  $\Delta S$  and the deceleration the scene demands is between  $-0.37$  and  $+0.07$ .

An exam that rewarded responsiveness alone would therefore rank first the policy with the most sub-1.5 s encounters. A score merges these three readings into one number. The edited frames keep them apart.

### C. The Five Exams

Table II gives the six profiles, read under our open-loop protocol on the 236 near-pedestrian nuScenes frames: the behaviour behind the ranks of Section IV-E.

a) Exposure: The six policies do not even agree on how far to plan, and this alone will turn out to explain more of the collision picture below than perception does: exposure spans a factor of four, from DiffusionDrive at half

the human’s distance to SimLingo at more than twice it, with AutoVLA and Alpamayo-1.5 near the human’s own.

b) Hazard sensitivity: On these frames, no policy’s plan responds to the pedestrian: on units that need a reaction, seeing it buys only 0.00–0.10 of safety on a  $[-1, 1]$  scale, against 0.57 when the logged human trajectory  $H$  takes the place of  $X^O$  in (7) (Table II). The benefit is also unreliable in direction: counting only cases where seeing the pedestrian changes the outcome, it turns a collision into a clean pass about as often as it turns a pass into a collision, for every policy (DiffusionDrive 9 vs. 5, SimLingo 18 vs. 30, the rest at most 5 either way, all McNemar  $p \geq 0.11$ ) — what reads as a safety margin in Table II does not carry over to the individual case.

c) Scaling: The benefit does not grow with the hazard: its rank correlation with  $a_{\text{req}}$  is negative for 5 policies, down to  $-0.28$  for DiffusionDrive. DiffusionDriveV2 is the only exception ( $+0.23$ ), and its benefit stays below 0.15. Replacing the hazard axis by the time-based  $\text{TTC}_0$ , whose sign is reversed by construction, leaves the picture unchanged ( $-0.07$  to  $+0.32$ , against  $-0.28$  to  $+0.23$ ). The human reference is  $+0.13$ : a driver who already yields at every hazard level has little room left to scale, so this exam is informative where HS is small, which is where all six policies sit.

d) Specificity: A policy whose plan moves when nothing requires it is not cautious but unsteady, and steadiness is what turns out to travel across a domain shift (Section IV-E). SimLingo’s plan moves most when nothing requires it ( $\text{SP} = 0.44$ ), while the other five stay steadier, at 0.65–0.84.

e) Lighting: Removing the pedestrian moves every plan less than the night-style perturbation does ( $\text{CFR} < 1$  for all six, Table II). Section IV-D reads this exam in depth.

f) Collisions: Collisions with the pedestrian’s logged future follow how far a policy plans to drive, not what it perceives. At the logged speed, the visible and removed rates differ by at most 0.4 points for every policy, and at 8 m/s by at most 3.8. SimLingo, which plans the longest trajectories, collides most. The low-exposure policies almost never reach the pedestrian. DiffusionDrive is safe because it stays short, AutoVLA and Alpamayo-1.5 cover a human-like distance without yielding, and SimLingo covers the most ground with the lowest specificity.

g) Why the risk goes unmanaged: The check-up only diagnoses. Two mechanistic observations on the three BEV planners explain the diagnosis without altering it. First, the planners’ own BEV semantic heads do not decode the pedestrian: on 520 NAVSIM near-pedestrian scenes the pedestrian channel is no brighter at the pedestrian than at a mirrored empty location, and never wins the argmax inside a pedestrian’s box, although the pedestrian does leave a steerable trace in the features. Second, the plan does not reach the pedestrian: plans end 5–9 m ahead while the median corridor pedestrian stands 20 m away,

TABLE II

Diagnostic profiles. Brackets: 95% bootstrap CI. Arrows: outside the reference threshold of Table I, pointing to the side the value falls on; bold: best per column. Higher is better for hazard sensitivity, scaling, specificity and lighting, lower for collision; exposure is read against the human’s distance. Collision:  $1 - A(X)$  (5) at 8 m/s,  $X^O / X^R$ . Human: the logged trajectory  $H$  in place of  $X^O$ , scored against each policy’s blind plan; HS averaged over the six pairings, scaling their median. Specificity and lighting need a counterfactual query and have no human counterpart.

| Policy         | Exposure<br>Exp    | Hazard sens.<br>HS  | Scaling<br>Sc         | Specificity<br>SP  | Lighting<br>CFR    | Collision<br>$O/R$ , 8 m/s |
|----------------|--------------------|---------------------|-----------------------|--------------------|--------------------|----------------------------|
| DD             | 0.50↓ [0.46, 0.55] | 0.06↓ [-0.01, 0.14] | -0.28↓ [-0.50, -0.05] | 0.76↓ [0.73, 0.80] | 0.23↓ [0.17, 0.30] | 7.6 / 9.2                  |
| LTF            | 0.79↓ [0.75, 0.84] | 0.03↓ [0.00, 0.06]  | -0.13↓ [-0.34, +0.07] | 0.78↓ [0.75, 0.81] | 0.23↓ [0.19, 0.28] | 9.9 / 10.7                 |
| DDv2           | 0.88 [0.77, 0.95]  | 0.10↓ [0.05, 0.16]  | +0.23 [-0.02, +0.50]  | 0.65↓ [0.61, 0.70] | 0.47↓ [0.37, 0.58] | 3.1 / 6.1                  |
| SimLingo       | 2.19↑ [1.98, 2.65] | 0.00↓ [-0.03, 0.03] | -0.22↓ [-0.38, -0.08] | 0.44↓ [0.38, 0.50] | 0.23↓ [0.18, 0.29] | 40.5 / 36.6                |
| AutoVLA        | 1.01 [0.97, 1.11]  | 0.02↓ [-0.00, 0.05] | -0.18↓ [-0.34, +0.01] | 0.76↓ [0.72, 0.79] | 0.44↓ [0.33, 0.55] | 32.1 / 29.8                |
| Alpamayo 1.5   | 1.13 [1.06, 1.21]  | 0.01↓ [-0.01, 0.02] | -0.23↓ [-0.49, +0.03] | 0.84 [0.81, 0.87]  | 0.25↓ [0.16, 0.35] | 11.2 / 11.2                |
| Human (logged) | 1.00               | 0.57                | +0.13                 | —                  | —                  | —                          |

TABLE III

The night-style perturbation and the plan. CFR =  $\mathbb{E}F/\mathbb{E}I$  (10) on the same frames, 95% CI:  $F$  (pedestrian) should be large,  $I$  (re-lighting) small, so CFR higher is better. Speed: change of planned mean speed,  $X^O \rightarrow X^N$ .  $\Delta_N S = S(X^N) - S(X^O)$ ,  $S$  of (5), negative = closer. Last column: CFR at the dusk and night levels on the 246 NAVSIM scenes. Bold:  $p < 0.01$ .

| Policy       | CFR [95% CI]      | ↑ speed ↓ $\Delta_N S$ (m) | ↑ dusk / night CFR ↑ |
|--------------|-------------------|----------------------------|----------------------|
| DD           | 0.30 [0.20, 0.43] | <b>+37%</b> <b>-0.37</b>   | 0.52 / 0.35          |
| LTF          | 0.36 [0.24, 0.49] | <b>+12%</b> <b>-0.24</b>   | 0.34 / 0.15          |
| DDv2         | 0.49 [0.33, 0.70] | <b>-8%</b> <b>+0.15</b>    | 0.61 / 0.38          |
| SimLingo     | 0.31 [0.21, 0.44] | <b>+16%</b> <b>-0.47</b>   | 0.87 / 0.20          |
| AutoVLA      | 0.38 [0.21, 0.57] | -0.3% +0.005               | 0.45 / 0.26          |
| Alpamayo 1.5 | 0.18 [0.10, 0.30] | -2% +0.03                  | 0.54 / 0.19          |

and raising the input speed makes collisions rise identically with and without the pedestrian in view (Table II, visible against removed at 8 m/s). The limit is geometric before it is perceptual.

h) Reading the profiles: Read one exam at a time, the profiles answer separate questions that a single score merges. Whether a policy is safe because it yields or because it stays short is answered by exposure together with hazard sensitivity: DiffusionDrive keeps its distance by planning short trajectories, which holds only as long as the trajectories stay short, whereas AutoVLA and Alpamayo-1.5 cover a human-like distance without the response to match. Whether a plan that moves is a plan that yields is answered by the decomposition of Section IV-B.

#### D. The Lighting Exam in Depth

Lighting should not change what a driver does, so this exam asks how far a policy is moved by an irrelevant change. It is the most robust result in the check-up, and the verdict that transfers between driving sides for every policy (Section IV-E).

a) Magnitude: Removing a pedestrian from the corridor moves every policy’s plan only 0.18–0.49 times as far as the night-style perturbation does, with every 95% interval below one (Table III). On detector-verified edits CFR = 0.30–0.64, and only Alpamayo-1.5’s interval (45 frames) reaches above one.

b) Two lighting levels: The perturbation is applied at two strengths,  $\mathcal{E}_D$  (dusk) and  $\mathcal{E}_N$  (night), the milder dusk level chosen to keep the pedestrian visible. Of the 45 targets the detector finds in the original, it still finds 89% at dusk and 67% at the night level (removal: 2%). On the same 246 scenes the ratio is 0.34–0.87 at dusk and 0.15–0.38 at the night level: below one for every policy at both levels, with the pedestrian in plain view.

c) Direction: faster, and therefore closer: A policy that slows under the perturbation is doing something defensible, one that speeds up is not. Three policies plan to go faster under the night-style perturbation: DiffusionDrive by 37%, SimLingo by 16% and LTF by 12%, and their whole plans pass 0.24–0.47 m closer to the pedestrian. The approach is longitudinal: the further the plan’s endpoint moves forward, the closer it gets (rank correlation  $-0.73$  to  $-0.96$ ), while the endpoint shifts toward the pedestrian’s side no more often than chance. DiffusionDriveV2, the only policy that also reads LiDAR, slows ( $-8\%$ ) and keeps more distance. AutoVLA and Alpamayo-1.5 are unaffected. Whether these diagnostic profiles and policy orderings persist across substantial domain shifts is formally tested in Section IV-E.

#### E. Prognosis: What Transfers, and at What Cost

Two questions decide whether a few hundred edited frames can stand in for a rank: which verdicts carry over to the other driving side, and how many frames each verdict needs.

a) A pre-registered transfer test: Does a diagnosis made on one driving side describe the same policy on the other? Before computing any right-hand-drive statistic, we fixed six predictions, P1–P6 in Table IV, and their decision rules. They were derived from the profile of the 88 Boston frames, and we tested them on the near-pedestrian units of the 148 Singapore frames (Table IV). The registration covered the six policies public at that time. NVIDIA subsequently released Alpamayo 1.5, which accepts navigation instructions, and we use it in place of Alpamayo-R1 throughout this paper. Because both driving sides of Alpamayo 1.5 were measured together, its

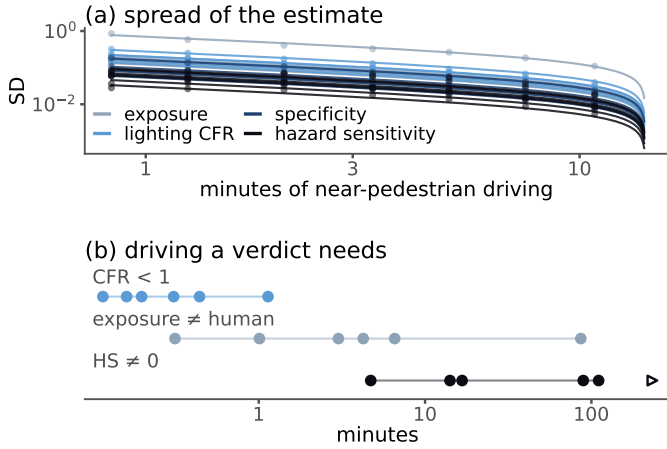


Fig. 3. What a verdict costs. One frame is 5 s of near-pedestrian driving. (a) Spread of each estimate against sample size with its fitted law, one line per policy. (b) Driving needed per verdict, one dot per policy, off-scale values at the axis end.

TABLE IV

Pre-registered transfer test. Predictions fixed on 88 Boston frames, tested on 148 Singapore frames; five policies, the replaced sixth changes no verdict (Section IV-E).

| No. | Prediction                                                               | Holds | Evidence          |
|-----|--------------------------------------------------------------------------|-------|-------------------|
| P1  | Lighting outweighs the pedestrian (CFR < 1) for every policy             | yes   | max upper CI 0.70 |
| P2  | Hazard sensitivity stays below 0.15 (upper CI)                           | no    | DD, DDv2 above    |
| P3  | Exposure ordering transfers ( $\rho \geq 0.6$ ; SimLingo top, DD bottom) | yes   | $\rho = 0.90$     |
| P4  | Specificity ordering transfers ( $\rho \geq 0.6$ ; SimLingo lowest SP)   | yes   | $\rho = 0.60$     |
| P5  | Point values stay within the left-hand CI ( $\geq 70\%$ of 15 cells)     | no    | 60%               |
| P6  | SimLingo collides most, visible vs. removed within 3 pts                 | yes   | gap 2.7 pts       |

numbers do not satisfy the registration discipline, so the table is evaluated on the five policies that do. Retaining the replaced policy changes none of the six verdicts.

b) What transferred: 4 of the six predictions hold, and they are the orderings and verdicts. The night-style perturbation outweighs the pedestrian in every policy. Exposure keeps its order ( $\rho = 0.90$ ), and so does specificity, with SimLingo the lowest. SimLingo collides most, with the visible and removed rates within 3 points of each other.

c) What did not: Two predictions fail.

- Point values drift: only 60% of the model-exam cells stay inside their Boston interval.
- In Singapore, DiffusionDrive and DiffusionDriveV2 show a small but non-zero hazard sensitivity (HS = 0.09 and 0.08).

d) Reading: A small check-up can therefore report an ordering and a verdict for use in a new domain, but not a calibrated value.

e) The standard scores on the same frames: These policies are usually compared by a leaderboard rank or an open-loop error. Running those scores and the check-up on the same six policies (Table V) lets us ask what

TABLE V

The six policies under the standard scores. Left: nuScenes open-loop L2 on the 236 near-pedestrian frames, original/ removed. Middle: NAVSIM EPDMS on all 783 Singapore scenes and on the 260 near-pedestrian ones. Right: share of moving-ego scenes with pedestrian TTC < 1.5 s per driving side, rank in parentheses.  $\uparrow/\downarrow$ : higher/lower is better. Bold: best per column.

|              | L2 (m) ↓    | EPDMS ↑   |           | TTC<1.5 s (%) ↓ |          |
|--------------|-------------|-----------|-----------|-----------------|----------|
| Policy       | orig. / rm. | all       | near      | LHD             | RHD      |
| DD           | 0.98 / 0.99 | 0.535 (4) | 0.744 (2) | 3.2 (1)         | 14.1 (1) |
| LTF          | 0.88 / 0.91 | 0.509 (5) | 0.735 (3) | 5.4 (2)         | 16.5 (2) |
| DDv2         | 1.35 / 1.31 | 0.652 (3) | 0.367 (6) | 18.3 (3)        | 22.4 (5) |
| SimLingo     | 3.28 / 3.35 | 0.359 (6) | 0.583 (4) | 61.3 (6)        | 56.5 (6) |
| AutoVLA      | 0.68 / 0.66 | 0.683 (1) | 0.413 (5) | 19.4 (4)        | 21.2 (4) |
| Alpamayo 1.5 | 0.77 / 0.72 | 0.663 (2) | 0.760 (1) | 25.4 (5)        | 20.0 (3) |

each recovers that the other does not, and which of them predicts behaviour at a site it has not seen.

f) nuScenes open-loop metrics: On our 236 near-pedestrian frames, removing the pedestrian from the image changes none of the standard open-loop numbers: the L2 error by at most 0.07 m, the frontal collision rate by at most 1.7 points and the pedestrian collision rate by at most 3.0 (Table V shows the L2 pairs, the two collision rates behave the same way and are omitted). The metric cannot tell whether a policy saw the pedestrian. Its ordering follows how far a policy's planned distance departs from the human's (rank correlation +0.66 with  $|\text{exposure} - 1|$ ): L2 rewards driving at the human's speed. We exclude rear-end contacts from the collision count. The logged traffic does not react, so a policy that plans to drive slower than the human is scored as being hit from behind.

g) NAVSIM, sliced to near pedestrians: On all 783 Singapore scenes, AutoVLA, Alpamayo-1.5 and DiffusionDriveV2 score highest. On the 260 scenes with a pedestrian near the ego corridor, the ordering is unrelated to that one ( $\rho = -0.09$ ). The two rank columns of Table V can be read side by side: the leaderboard's first (AutoVLA) falls to fifth and its third (DiffusionDriveV2) to last, while its fourth and fifth (DiffusionDrive, LTF) rise to second and third. The two orderings therefore select different policies. In these scenes the no-collision gate carries 100% of the spread, while over all 783 scenes it carries 1% and the two map-compliance gates carry 95% (drivable area 76%, driving direction 19%): the leaderboard ranks these policies by map compliance, and none of the exams above is visible in it. Of our exams, specificity predicts the near-pedestrian ordering ( $\rho = +0.66$ ), while hazard sensitivity does not (+0.14), because it is near zero for every policy on these frames.

h) Pedestrian TTC across driving sides: Following the plan against the pedestrian's logged future, we count the moving-ego scenes in which the minimum time to collision falls below 1.5 s. Near-pedestrian scenes from the left-hand-drive cities (nuScenes Boston and NAVSIM Las Vegas, Boston, Pittsburgh) and from Singapore give orderings that agree at  $\rho = +0.77$ . Both orderings coincide with planned distance ( $\rho = +1.00$  and +0.77 with exposure).

NAVSIM’s own TTC term, which counts all objects over 4s, agrees only weakly (0.31–0.54).

i) Which exam travels: This is the question a deployment decision has to answer. The target is the ordering of the six policies on the Singapore near-pedestrian scenes (original images), by TTC violations, clearance to the pedestrian, pedestrian collisions and EPDMS. Each predictor is an ordering obtained without that data. The standard evaluations do not predict it: nuScenes L2 and collision rates, and the NAVSIM leaderboard score, correlate with it at  $-0.54$  to  $+0.49$ . The same pedestrian-centred readouts taken on Boston or on the US cities do (0.83–1.00), and so does the exposure exam (0.94). Near-pedestrian EPDMS is predicted best by near-pedestrian EPDMS elsewhere (0.89) and by specificity (0.77). Hazard sensitivity and lighting invariance measure something else and do not predict these orderings. Of the five exams, the one whose ordering best matches how often a policy keeps clear of the pedestrian on the new side (share of scenes with minimum time-to-proximity  $\geq 1.5$ s, the complement of the TTC column of Table V) is not the one that measures how much a policy reacts but the one that measures how steadily it drives: over the six policies and the 246-scene pool of Section IV-B, specificity gives  $\rho = +0.94$ , against  $-0.77$  for exposure and  $-0.37$  for CFR (higher is better for all three exams, so a positive  $\rho$  means the exam’s ordering carries over). This exam was not pre-registered, and we report it as exploratory. Cross-policy correlations rest on six policies, and we read them as orderings, not as tests. Specificity asks how far a plan moves when an irrelevant part of the scene is edited. A policy that answers “not at all” drives steadily, and steadiness is what survives the change of side. The decomposition of Section IV-B says why: displacement that is not aimed away from the pedestrian is not protection, and a policy producing a lot of it comes close to the pedestrian on both sides. Where a standard metric measures what an exam measures, they agree (pedestrian collisions at  $\rho = +0.94$ , TTC and exposure). None shows whether the pedestrian changed the plan, and the full-set score hides the near-pedestrian scenes. A rank does not carry the near-pedestrian ordering to a new site. The exposure and specificity exams, read from the same frames, do.

j) What a verdict costs: For every exam and policy we subsample  $n$  of the 170 frames that all six policies share, 400 times, and fit the spread of the estimate with  $\text{sd}(n) = a\sqrt{1/n - 1/M}$ ,  $M = 170$  (Fig. 3).

k) Well-fitting exams: Exposure, the lighting ratio and specificity obey this law ( $R^2 = 0.97$ –1.00). The fitted  $a$  converts directly into the number of frames a verdict needs.

- Lighting outweighs the pedestrian (CFR  $< 1$ ) is settled for every policy within 14 frames.
- Exposure differs from the human’s is settled within 51 frames for the four policies whose exposure is far

from one. For AutoVLA and Alpamayo-1.5 it would take hundreds to thousands of frames, which is itself a finding, since they plan a human-like distance.

- Orderings of the six policies stabilise quickly: exposure with 10 frames (96% of subsamples reach  $\rho \geq 0.6$  with the full ordering) and specificity with 40 (99%).

l) Poorly fitting exams: Hazard sensitivity behaves differently. Its fits are poor ( $R^2$  as low as 0.26) wherever few units need a reaction. Deciding that HS  $\neq 0$  needs 57–1,326 frames, except for SimLingo, whose estimate sits on the threshold, and the ordering reaches 84% stability only at 90 frames.

In wall-clock terms, a 20s scene in this corpus yields about 4 usable near-pedestrian frames, so the lighting verdict costs roughly 3 scenes and the exposure verdict 13. Deciding that hazard sensitivity differs from zero costs 331—about 110 minutes of logged driving that actually contains a pedestrian near the corridor.

The division is therefore clear. A few dozen frames diagnose exposure, specificity and lighting. Whether a policy manages hazard needs hundreds of frames that contain a genuine need to react.

m) What a small sample in the new domain buys: We draw  $k$  scenes at random from the NAVSIM pool of Section IV-B, re-measure  $F$  and  $I$  on them and compare against the large-sample values, 200 draws per  $k$ . Ordering and level behave very differently. The ordering of how often the policies come close—the share of scenes whose minimum time-to-proximity falls below 1.5s—is recovered almost immediately:  $\rho = +0.64$  at  $k = 10$  and  $\rho = +0.65$  at  $k = 20$  ( $\rho \geq 0.6$  in 70% of draws), rising to  $\rho = +0.76$  at  $k = 40$ . The level is not: the same  $k = 20$  sample misstates the violation rate by 13 points and CFR by 0.116, and the rule hit-rates sit in between, at 3.6 points. Importing the Boston CFR values instead misstates the Singapore ones by 0.099, already worse than re-measuring  $k = 40$  fresh scenes (0.078), while 80 scenes reach 0.054 and 160 0.027. Values measured elsewhere are worth about forty scenes measured here.

## V. Discussion

a) Limitations: The exams are open-loop. The approach sequences are replayed from logs and re-anchored to the logged pose. The night-style perturbation is photometric, not a captured night scene, and it also halves the number of other pedestrians the detector finds. The counterfactual pairs are constructed by editing, not captured, and so carry noise: 8.5% of removals leave a detectable residue, the inpainting artefact is 5–17% of the effect, and the VLA models can be edited only through their front camera. Alpamayo-1.5 covers 170 of the 236 frames. SimLingo plans 2.5s, so for NAVSIM’s 4s we extrapolate at constant velocity (holding position instead gives EPDMS 0.47, still the lowest). Cross-policy correlations rest on six policies.

b) Next steps: The check-up is the first half of a benchmark-to-deployment loop, and the sample-size laws above size the second half. First, the paired data will be captured rather than constructed: the same scene with and without a pedestrian, recorded on our vehicle and re-rendered with 3D Gaussian splatting, so that a real-vehicle pair replaces the inpainted one. Second, the day-night pairs will move from a photometric transform to captured or physically rendered lighting, so that the lighting exam reads real shadows and exposure rather than a gain-and-gamma proxy. Third, the check-up will run inside a data flywheel: the VLA policies are post-trained on target-domain data and re-examined, so that the diagnosis measures what post-training recovers rather than a policy failing outright at the new site. The exams already price the collection: a few dozen pairs for exposure, specificity and lighting, hundreds with a genuine need to react for hazard sensitivity.

## VI. Conclusion

A benchmark score reports an outcome, not the behaviour behind it, so a rank cannot be trusted to hold at a new site. We proposed a check-up that reads that behaviour from a few hundred logged frames through verified, pixel-level counterfactual edits, reducing each scene to two axes of causal attribution—the faithfulness displacement  $F$  and the interference displacement  $I$ —on which five exams are built. Three results carry the paper. First, on the cells in which a pedestrian does lie on the planned path, only 1.9% of responses are genuine avoidance, and in open loop no policy’s plan diverts or slows, and a photometric night-style perturbation moves every plan further than the pedestrian does. Second, diagnosed on one driving side and tested on the other, orderings and verdicts carry over and point values do not. The exam whose ordering best matches how often a policy keeps clear of the pedestrian on the new side is specificity ( $\rho = +0.94$ ), where the standard scores reach only from  $-0.54$  to  $+0.49$ . Third, each verdict can be priced in advance—the lighting verdict within 14 frames, exposure within 51—so that a deployment decision costs tens of target-domain frames rather than a fleet test.

## References

- [1] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, “nuScenes: A multimodal dataset for autonomous driving,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [2] H. Caesar, J. Kabzan, K. S. Tan, W. K. Fong, E. Wolff, A. Lang, L. Fletcher, O. Beijbom, and S. Omari, “nuplan: A closed-loop ml-based planning benchmark for autonomous vehicles,” in *CVPR ADP3 Workshop*, 2021.
- [3] D. Dauner, M. Hallgarten, T. Li, X. Weng, Z. Huang, Z. Yang, H. Li, I. Gilitschenski, B. Ivanovic, M. Pavone, A. Geiger, and K. Chitta, “NAVSIM: Data-driven non-reactive autonomous vehicle simulation and benchmarking,” in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024.
- [4] S. Z. Zhao, L. Wang, H. Ruan, Y. Bao, Y. Chen, Z. Leng, A. Ravichandran, H. He, Z. Zhou, X. Han, A. Peri, Z. Huang, P. Desai, H. Christensen, J. Ma, and B. Zhou, “BridgeSim: Unveiling the OL-CL gap in end-to-end autonomous driving,” *arXiv preprint arXiv:2604.10856*, 2026.
- [5] Z. Li, Z. Yu, S. Lan, J. Li, J. Kautz, T. Lu, and J. M. Alvarez, “Is ego status all you need for open-loop end-to-end autonomous driving?” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [6] F. Codevilla, E. Santana, A. M. López, and A. Gaidon, “Exploring the limitations of behavior cloning for autonomous driving,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- [7] K. D. Kusano, J. M. Scanlon, Y.-H. Chen, T. L. McMurtry, T. Gode, and T. Victor, “Comparison of Waymo rider-only crash rates by crash type to human benchmarks at 56.7 million miles,” *Traffic Injury Prevention*, vol. 26, no. S1, pp. S8–S20, 2025.
- [8] Physical Intelligence, A. Amin, R. Aniceto, A. Balakrishna, K. Black, K. Conley, G. Connors, J. Darpinian, K. Dhabalia, J. DiCarlo, D. Driess, M. Equi, A. Esmail, Y. Fang, C. Finn, C. Glossop, T. Godden, I. Goryachev, L. Groom, H. Hancock, K. Hausman, G. Hussein, B. Ichter, S. Jakubczak, R. Jen, T. Jones, B. Katz, L. Ke, C. Kuchi, M. Lamb, D. LeBlanc, S. Levine, A. Li-Bell, Y. Lu, V. Mano, M. Mothukuri, S. Nair, K. Pertsch, A. Z. Ren, C. Sharma, L. X. Shi, L. Smith, J. T. Springenberg, K. Stachowicz, W. Stoeckle, A. Swerdlow, J. Tanner, M. Torne, Q. Vuong, A. Walling, H. Wang, B. Williams, S. Yoo, L. Yu, U. Zhilinsky, and Z. Zhou, “ $\pi_{0.6}^*$ : A VLA that learns from experience,” *arXiv preprint arXiv:2511.14759*, 2025.
- [9] N. Kalra and S. M. Paddock, “Driving to safety: How many miles of driving would it take to demonstrate autonomous vehicle reliability?” *RAND Corporation, Tech. Rep. RR-1478*, 2016.
- [10] Y. Ganin and V. Lempitsky, “Unsupervised domain adaptation by backpropagation,” in *International Conference on Machine Learning (ICML)*, 2015.
- [11] I. Gulrajani and D. Lopez-Paz, “In search of lost domain generalization,” in *International Conference on Learning Representations (ICLR)*, 2021.
- [12] M. Arjovsky, L. Bottou, I. Gulrajani, and D. Lopez-Paz, “Invariant risk minimization,” *arXiv preprint arXiv:1907.02893*, 2019.
- [13] N. Kachaev, M. Kolosov, D. Zelezetsky, A. K. Kovalev, and A. I. Panov, “Don’t blind your vla: Aligning visual representations for ood generalization,” *arXiv preprint arXiv:2510.25616*, 2025.
- [14] A. Zou, L. Phan, S. Chen, J. Campbell, P. Guo, R. Ren, A. Pan, X. Yin, M. Mazeika, A.-K. Dombrowski et al., “Representation engineering: A top-down approach to AI transparency,” *arXiv preprint arXiv:2310.01405*, 2023.
- [15] E. Todd, M. L. Li, A. S. Sharma, A. Mueller, B. C. Wallace, and D. Bau, “Function vectors in large language models,” *arXiv preprint arXiv:2310.15213*, 2024. [Online]. Available: <https://arxiv.org/abs/2310.15213>
- [16] K. Chitta, A. Prakash, B. Jaeger, Z. Yu, K. Renz, and A. Geiger, “TransFuser: Imitation with transformer-based sensor fusion for autonomous driving,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023.
- [17] B. Liao, S. Chen, H. Yin, B. Jiang, C. Wang, S. Yan, X. Zhang, X. Li, Y. Zhang, Q. Zhang, and X. Wang, “DiffusionDrive: Truncated diffusion model for end-to-end autonomous driving,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [18] J. Zou, S. Chen, B. Liao, Z. Zheng, Y. Song, L. Zhang, Q. Zhang, W. Liu, and X. Wang, “DiffusionDriveV2: Reinforcement learning-constrained truncated diffusion modeling in end-to-end autonomous driving,” *arXiv preprint arXiv:2512.07745*, 2025.
- [19] K. Renz, L. Chen, E. Arani, and O. Sinavski, “SimLingo: Vision-only closed-loop autonomous driving with language-action alignment,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [20] Z. Zhou, T. Cai, S. Z. Zhao, Y. Zhang, Z. Huang, B. Zhou, and J. Ma, “AutoVLA: A vision-language-action model for end-to-end autonomous driving with adaptive reasoning and

- reinforcement fine-tuning,” arXiv preprint arXiv:2506.13757, 2025.
- [21] NVIDIA, “Alpamayo-R1: Bridging reasoning and action prediction for generalizable autonomous driving in the long tail,” arXiv preprint arXiv:2511.00088, 2025.
  - [22] —, “Alpamayo 1.5: An open reasoning vision-language-action model with navigation guidance,” 2026, model card and code, <https://github.com/NVlabs/alpamayo1.5>.
  - [23] W. Cao, M. Hallgarten, T. Li, D. Dauner, X. Gu, C. Wang, Y. Miron, M. Aiello, H. Li, I. Gilitschenski, B. Ivanovic, M. Pavone, A. Geiger, and K. Chitta, “Pseudo-simulation for autonomous driving,” in *Conference on Robot Learning (CoRL)*, 2025.
  - [24] B. Jaeger, K. Chitta, and A. Geiger, “Hidden biases of end-to-end driving models,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
  - [25] X. Jia, Z. Yang, Q. Li, Z. Zhang, and J. Yan, “Bench2Drive: Towards multi-ability benchmarking of closed-loop end-to-end autonomous driving,” in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024.
  - [26] M. T. Ribeiro, T. Wu, C. Guestrin, and S. Singh, “Beyond accuracy: Behavioral testing of NLP models with CheckList,” in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.
  - [27] D. Hendrycks and T. Dietterich, “Benchmarking neural network robustness to common corruptions and perturbations,” in *International Conference on Learning Representations (ICLR)*, 2019.
  - [28] L. Kong, Y. Liu, X. Li, R. Chen, W. Zhang, J. Ren, L. Pan, K. Chen, and Z. Liu, “Robo3D: Towards robust and reliable 3D perception against corruptions,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
  - [29] S. Xie, L. Kong, W. Zhang, J. Ren, L. Pan, K. Chen, and Z. Liu, “RoboBEV: Towards robust bird’s eye view perception under corruptions,” arXiv preprint arXiv:2304.06719, 2023.
  - [30] P. de Haan, D. Jayaraman, and S. Levine, “Causal confusion in imitation learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
  - [31] C. Wen, J. Lin, T. Darrell, D. Jayaraman, and Y. Gao, “Fighting copycat agents in behavioral cloning from observation histories,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
  - [32] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.
  - [33] P. Jacob, É. Zablocki, H. Ben-Younes, M. Chen, P. Pérez, and M. Cord, “STEEX: Steering counterfactual explanations with semantics,” in *European Conference on Computer Vision (ECCV)*, 2022.
  - [34] M. Zemni, M. Chen, É. Zablocki, H. Ben-Younes, P. Pérez, and M. Cord, “OCTET: Object-aware counterfactual explanations,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
  - [35] R. Roelofs, L. Sun, B. Caine, K. S. Refaat, B. Sapp, S. Ettinger, and W. Chai, “CausalAgents: A robustness benchmark for motion forecasting using causal relationships,” arXiv preprint arXiv:2207.03586, 2022.
  - [36] A. Azaria and T. Mitchell, “The internal state of an LLM knows when it’s lying,” in *Findings of the Association for Computational Linguistics: EMNLP*, 2023.
  - [37] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, and R. Girshick, “Segment anything,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
  - [38] R. Suvorov, E. Logacheva, A. Mashikhin, A. Remizova, A. Ashukha, A. Silvestrov, N. Kong, H. Goka, K. Park, and V. Lempitsky, “Resolution-robust large mask inpainting with fourier convolutions,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2022.
  - [39] S. Ren, K. He, R. Girshick, and J. Sun, “Faster R-CNN: Towards real-time object detection with region proposal networks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
  - [40] E. T. Hall, *The Hidden Dimension*. Doubleday, 1966.
  - [41] D. Gettman and L. Head, “Surrogate safety measures from traffic simulation models,” *Transportation Research Record*, vol. 1840, no. 1, pp. 104–115, 2003.
  - [42] J. C. Hayward, “Near-miss determination through use of a scale of danger,” in *Highway Research Record*, no. 384, 1972, pp. 24–34.
  - [43] T. Kruse, A. K. Pandey, R. Alami, and A. Kirsch, “Human-aware robot navigation: A survey,” *Robotics and Autonomous Systems*, vol. 61, no. 12, pp. 1726–1743, 2013.
  - [44] P. Hase, M. Bansal, B. Kim, and A. Ghandeharioun, “Does localization inform editing? surprising differences in causality-based localization vs. knowledge editing in language models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.